Project Vision Document

01 — Project Idea & Vision Statement

1.1 Project Name: Note Taking Assistant

1.2 One-Line Pitch:

An assistant that helps you take notes alongside you so that you can keep pace with the lecture currently going on.

1.3 Problem Statement:

Students taking notes during a lecture have to split their attention between listening and writing. Focusing on one usually costs them the other. Even attentive students have brief lapses in attention and come back having missed an explanation, an example, or a warning about a common mistake. Research on note-taking shows how large these gaps can be. Students watching a lecture once recorded about 80% of its main ideas but only around 35% of its supporting details (Kiewra et al., 1991, as cited in Luo et al., 2016). The result is notes with gaps the student often isn't aware of until they're studying for an exam.

Existing tools don't solve this well. Lecture recordings and transcripts contain everything, but searching an hour of material for a few missed points is impractical. AI summarizers produce a separate set of notes, which replaces the student's own note-taking instead of improving it. That's a problem, because revising their own notes is itself valuable. Luo et al. (2016) found that students who revised their notes recorded more complete notes and scored higher on later tests.

1.4 Proposed Solution:

The system takes three inputs: the lecture audio, the slides, and the student's notes. The audio is transcribed with Whisper, and text is pulled from the slides and notes using OCR (or read directly if they're already digital). The transcript is then split into sections based on which slide was on screen, so each part of the lecture can be matched to the notes that cover it.

To check what the student captured, the system compares their notes to each section using Sentence Transformers embeddings. This means a student can paraphrase the professor and the system will still count that content as covered.

Not everything missing is worth flagging, so the system also scores importance. Signals include phrases like "this will be on the exam" or "a common mistake is…", content that appears on

both the slide and in the spoken lecture, and content types such as definitions, equations, and examples.

The output is a list of missed content grouped by slide, with timestamps so the student can jump back to that part of the lecture. The system does not write notes for the student. It only flags gaps, leaving the student to fill them in themselves.

Beyond the core gap detection, several extensions are planned as stretch goals. The system could generate practice quizzes from the content a student missed, then use their quiz results to point them back to the lecture sections they need to review. A professor-facing view could combine anonymized gap data across a class, showing which parts of a lecture students most often miss. The same alignment approach could also check whether an assignment's content matches what was covered in lecture. These features build on the core system but are not required for it to work.

1.5 Vision Statement:

We want to close the gap between what's taught and what's written down. For students, that means knowing exactly what their notes are missing and filling it in themselves. For professors, it means seeing which parts of a lecture students struggle to capture, and using that to improve how they teach.

02 — Target Users & Stakeholders

2.1 Primary User Persona:

An Undergraduate College/University Student majoring in a lecture heavy programs like engineering/sciences possibly working a part time job as well.

The student attends multiple lectures a week and takes notes on the computer/tablet. In a fast paced lecture, they may miss important explanations, examples, or concepts when trying to keep up with the professor or they may lose focus and struggle trying to figure out what they missed. Before exams or quizzes when reviewing their notes they may not have enough time to rewatch an entire lecture.

During a live lecture, the students can record audio through the application while continuing to take their own notes. The application assistant helps them keep their notes complete by showing important information, explanations, equations, examples, or code they may have missed as the lecture moves on. If the student falls behind there will be a catch up feature that they can use to see what was covered from that point so that they can continue from the current point in the lecture. After the lecture the full gap detection system then compares the lecture, slides, and the students notes to find any important information that the student may have missed and where it came from in the lecture. The students will also be able to use the application to create practices quizzes, coding support, help with preparation for exams, visualization help for mathematical problems, and other support options in the application.

2.2 Secondary User Persona:

A Course Instructor - Teaches multiple courses with 80+ students
The professor would like their students to understand the important information being taught during lectures, however they might not always know which topics students are struggling with understanding or commonly missing in their notes.

As future work, there can be a professor side of the application where they can see class wide learning gaps. This can show the professor which concepts the students are commonly struggling with. This can help the professor know which topics may need to be explained again or taught in a different way.

Professors can also provide course materials such as lecture slides, learning objectives, assignments, and rubrics to help the application better understand what is being taught in the course and provide students with more accurate and personalized support.

2.3 Stakeholders list:

Stakeholder	Interest
College/University Student	Primary users use the application for live note assistance, exam prep, assignment analysis, coding support, as well as other learning support tools.
Instructor	Professors provide the lecture content that the students learn from and in the future development could use class wide learning insights with the development of a professor facing dashboard.
Colleges/Universities	Could integrate the platform as a learning/academic support tool for their students.
Student Support and Accessibility Services	Benefit from the application as an extra tool for students who have difficulties keeping up with lectures or focus.
Development Team	Responsible for developing, testing, improving and maintaining the system and its models.

2.4 User Pain Points:

Some students struggle to listen and understand while also taking notes at the same time often causing them to miss some important information during the lectures. They can also fall behind and lose focus not realizing what is missing from their notes until they start studying. The note taking assistant helps students keep up during lectures, fill in missing information in notes, and prepare for quizzes, exams, and assignments.

2.5 Accessibility & Equity Considerations:

Students have varying styles of learning and may process information at different speeds. This application will provide clear text, keyboard navigation, transcripts, and visual learning options to make sure the content is easy to understand and access.Also because background noise, accents, or poor audio quality could affect the transcript the application will let students know when the audio may not have been understood correctly.

03 — Market Analysis & Competitive Landscape

3.1 Total Addressable Market (TAM)

The primary market for this product is college and university students especially students in lecture heavy programs or have different learning needs.

In Canada, there are over 2 million students enrolled in colleges and universities every year creating a large potential market for the Note Taking Assistant. The majority of those students attend heavy lecture based courses and could benefit from additional support with the note taking, understanding course materials, and studying.

This application would initially target post secondary students in Canada, however it could eventually expand later to students internationally, Colleges and universities could also become part of the market by providing/integrating the application with their school websites.

3.2 Competitive Analysis Table

Features	Note Taking Assistant	Notability	Quizlet	Otter.ai
Live Lecture Support (transcription, recording, and Catch Up)	✓-yes	partial	x	partial
Practice Quizzes/ Exam Prep	✓	partial	x	x
Math/Coding Learning Support	✓	partial	x	x
Assignment Analysis	✓	x	x	x
Chatbot for Question and Answer Support (Using Lectures and Notes)	✓	x	partial	x

3.3 Competitive Differentiator

The main difference between the Note Taking Assistant and other similar applications is it focuses on what the specific student using the application is struggling with rather than just creating notes, transcripts, or study materials. Note Taking Ai compares lectures, slide materials, and students notes to detect important information that students may have missed, and during a live lecture the application can help students keep their notes complete and use a catch up feature if they ever fall behind.

After the lecture, the information is put together and can be used to build a knowledge tracker that shows what the students understand and what they might need more help with. The Note Taking Assistant then takes these learning gaps and uses it to provide personalized practice quizzes/exams, math visualizations/explanations, assignment feedback, and coding support.

This makes the Note Taking Assistant different because the different features are connected and are able to adapt based on different student needs, rather than providing the same type of support to every student.

04 — Revenue & Business Model

4.1 Primary Revenue Stream

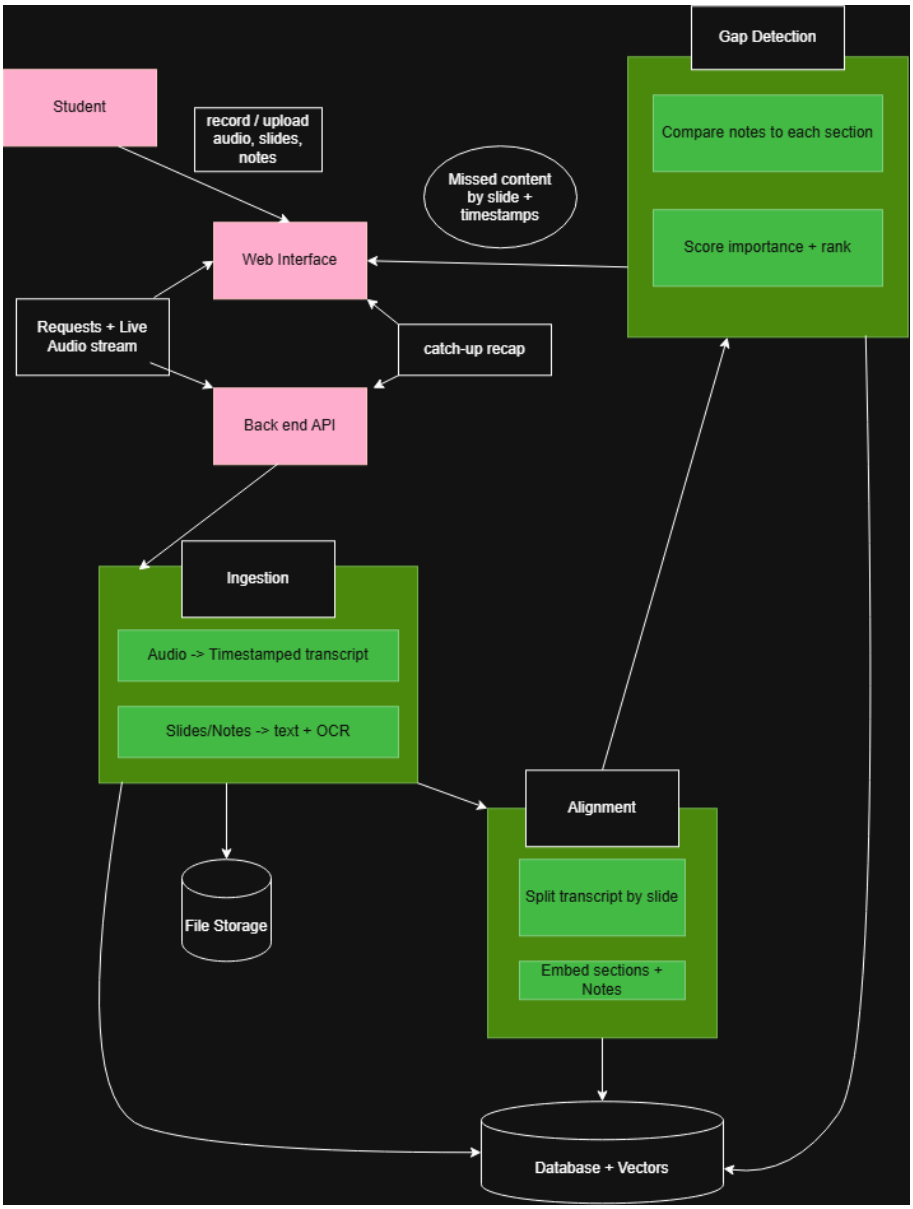
The note taking assistant will rely on a free and premium subscription model. Students will be able to use a free version of the application with access to basic features and a set number of lecture recordings/ Ai features each month. The free version will have advertisements which allow the application to earn revenue while still providing students with access to the main features without having to pay.Students will have the option to upgrade to a premium subscription plan with a monthly or yearly fee to remove advertisements and get full unlimited access to all features.

4.2 Secondary Revenue Opportunities

The application can also be offered directly to universities or colleges. Giving the schools an opportunity to purchase access for their students through a product license or yearly access. Also schools could pay to integrate this system with their learning platforms.

05 — Technical Stack & Architecture

5.1 Architecture Diagram



5.2 Data Layer

- A database for users, courses, lectures, transcript segments, slide text, note chunks, and flagged gaps
- Vector storage for embeddings, so similar text can be looked up quickly
- File storage for raw audio, slide files, and images of notes
- An evaluation dataset of 3–5 hand-labelled lectures, kept separate from anything used for tuning

5.3 ML / AI Layer

- Speech-to-text model that returns word-level timestamps (e.g. Whisper)
- Text extraction for digital slides and notes
- Slide alignment: compare slide text against the transcript to find when each slide was on screen
- Semantic similarity model that turns lecture sections and notes into embeddings, so a paraphrased point still counts as covered (e.g. Sentence Transformers)
- Importance Scoring based on verbal cues, overlap between slide and speech, and content type. It starts rule-based and may become a trained model later.
- Recap generation for catch-up that pulls the key sentences from the transcript since the student's chosen start time.

5.4 Backend / API Layer

- An API that handles uploads, starts analysis jobs, and returns results (e.g. FastAPI)
- A real-time connection for streaming audio during live recording
- A background job queue so long tasks like transcribing a full lecture don't block the app
- User authentication so students only see their own data / access FOL and Brightspace

5.5 Frontend / User Interface

- A web app that works in the browser and can access the microphone for recording
- Main screens:
    - Lecture list
    - Recording screen with a "catch up" button
    - Upload screen
    - Results screen showing missed content by slide, with a clickable timestamp
- Buttons for students to dismiss a flag or mark it as useful

5.6 Infrastructure & Cloud

- Services packaged in containers so they run the same on every team member's machine and on the server
- Development runs locally. The demo is hosted on a single cloud server with GPU access, likely paid for with student credits.

5.7 MLOps & Model Management

- An experiment tracker that records which models and settings were used and how each one scored
- Model versions pinned in a config file so results can be reproduced
- An evaluation script that runs the whole pipeline on the labelled lectures and reports the metrics from 6.5

5.8 CI/CD Pipeline

06 — Project Scope & Key Deliverables

6.1 In Scope:

- Transcribing lecture audio with Whisper, including timestamps
- Extracting text from slides (PDF/PPTX) and from typed notes
- OCR for handwritten or scanned notes
- Splitting the transcript into sections by slide
- Matching notes to lecture sections with Sentence Transformers embeddings, so paraphrased content counts as covered
- Scoring the importance of missed content using verbal cues (“this will be on the exam”), overlap between slide and speech, and content type (definitions, equations, examples)
- A FastAPI backend and a simple interface showing missed content by slide, with timestamps
- Recording lecture audio live through the app, then processing it after the lecture
- Showing hints or transcript exce
- Stretch goals
    - Practice quizzes built from the notes
    - Professor facing view of anonymized, class-wide gaps
- A "catch up" feature: the student marks when they stepped away or picks a start time, and the app shows a short recap of what was covered up to the current slide

6.2 Out of Scope

- Writing or auto-completing notes for the student
- Full real-time gap detection during a live lecture (gap detection runs after the lecture).

6.3 Sprint Milestone Map

Sprint	Focus	Milestone
1	Setup and data	Repo, environment, and a sample dataset of 3-5 lectures with audio, slides, and notes. Missed content labelled by hand for evaluation
2	Ingestion	Whisper transcription and slide/note text extraction working end to end
3	Alignment	Transcript split by slide, and notes matched to sections with embeddings
4	Gap Detection	Missed content identified and ranked by importance

5	Live recording and catch-up	In-app recording, transcription in short chunks, and a working catch-up recap
6	Interface, Evaluation and Polish	FastAPI backend and UI finished, metrics measured, stretch goals if time allows, final report and demo

6.4 Minimum Viable Product (MVP)

A student records a lecture in the app or uploads the audio, then provides the slides and their typed notes. After the lecture, the system returns a list of important content missing from the notes, grouped by slide, with timestamps pointing back to the recording. Paraphrased notes are recognized as covered.

During the lecture, the student can tap "catch up" to see a short recap of what was covered since they stepped away, up to the current slide. Recaps are shown on screen only and are not added to the student's notes.

Handwritten notes, practice quizzes, and the professor-facing view are stretch goals and not part of the MVP.

6.5 Success Metrics

- **Gap detection accuracy:** precision and recall against hand-labelled gaps in the test lectures (target: recall of at least 0.75)
- **Paraphrase matching:** percentage of paraphrased notes correctly counted as covered (target: 80%+)
- **Transcription quality:** Whisper word error rate on the test lectures (target: under 15%)
- **Processing time:** a one-hour lecture analyzed in under 10 minutes after it ends
- **Catch-up speed:** recap shown within 15 seconds of tapping "catch up"
- **Catch-up usefulness:** students rate whether the recap let them follow the lecture again (target: average of 4/5)
- **Overall user feedback:** a small group of students rating how useful the flagged gaps were (target: average of 4/5)

07 — Team Roles & Responsibilities

7.1 Team Member Profiles

Name	Background	Key skills
Matt Tennant	B.Sc. Computer Science (Western)	Python, Pytorch, NLP, deep learning, photoshop, audacity, model evaluations, classical ML
Rami Joukhaji	B.A.Sc. Electrical and Computer Engineering (University of Windsor)	Python, SQL, Machine Learning, Deep Learning, Computer Vision, Data Analysis, Model Training and Evaluation

08 — Ethical AI & Responsible Development

8.1 Data Bias Assessment

Accents and speech:

Whisper's accuracy varies by multiple variables such as accent, speaking speed, and audio quality. A professor with a strong accent or a noisy lecture hall means

Note-taking styles:

Students who use diagrams, heavy abbreviations, or a mix of languages may get flagged as "missing" content they actually captured.

Subject matter:

Sentence Transformers have been known to be weaker on equations, code, and chemical formulas than on plain text. STEM students, who are a part of the primary users, could receive more false flags because of this.

Teaching styles:

The importance score relies partly on verbal cues like "this will be on the exam." Professors who don't use phrases like that will have their key points ranked lower. Slide overlap and content type are included as signals so importance doesn't depend on one kind of professor

8.2 Model Explainability

Every flagged gap shows why it was flagged:

- The transcript excerpt and a timestamp link to hear it in context
- The closest match found in the student's notes and how similar it was, so the student can see the system looked and didn't find it.
- Which importance signals were triggered (e.g. "verbal cues)

Each importance rating relies on a weighted combination of these factors, making it straightforward to see which elements drove the score. Should we swap this for a learned classifier down the road, we will use an explainable architecture like logistic regression to keep feature weights transparent. Users can clear any flagged gap; the tool simply highlights potential omissions, leaving it up to the student to judge their relevance.

8.3 Privacy & Data Governance

Recording Consent:

Lecture recordings capture the professor and sometimes other students. The app will remind users to follow their institution's recording policy and get the instructor's permission before recording. Ontario colleges and universities fall under FIPPA, and if the product were ever offered commercially, PIPEDA would apply as well.

Data minimization:

- Raw audio can be deleted automatically once the transcript is created (on by default, with an option to keep it for timestamp playback).
- Only the data needed for gap detection is stored.

Ownership and control:

- Students own their notes and results and can export or permanently delete them at any time.
- Data is never used to train or tune models unless the student opts in.

Security:

- Files and database are encrypted at rest, and all traffic uses HTTPS.
- Each student can only access their own data.